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Cones, Pyramids, and Spheres

Toolbox

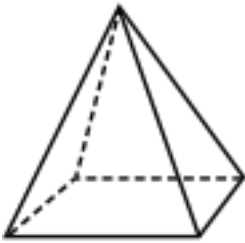
Cones, Pyramids, and Spheres

Every solid figure has volume. Remember, when you find volume, you are finding out how many cubes of a certain size would fit into the figure. Some shapes would be impossible to fill with cubes, so you need to use a formula to find the volume.

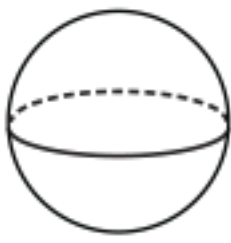
A **pyramid** has a base with straight sides and triangular faces that meet at a single vertex. The most common pyramid has a square base, but the base can be any polygon.

A **cone** has a circle for a base and one vertex.

A **sphere** is a three-dimensional object where each point on the surface is an equal distance from the center.







pyramid

cone

sphere

The GED math formula sheet has the formulas for finding the volumes of these figures. You don’t need to memorize them, but take time to understand how they work.

know how to use these formulas!

Volume (*V*) Formulas

Pyramid

$V = \frac{1}{3} Bh$, where B = area of the Base and h = height

Cone

$V = \frac{1}{3} \pi r^2 h$, where r = radius and h = height

Sphere

$V = \frac{4}{3} \pi r^3$, where r = radius

To solve volume problems, substitute the given values and solve. Use 3.14 for pi, or the π key on your calculator. On drawings involving height, the height is the perpendicular distance from the base to the vertex.

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Lesson 4: Volume

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